17.2. IRRADIATION YIELDS

The production of a radioactive nuclide by irradiation in a reactor or an accelerator can be written as:

$$X_{target} + X_{proj.} \xrightarrow{k} X_1 \xrightarrow{\lambda_1} X_2 \xrightarrow{\lambda_2} X_3 \tag{17.5}$$

We are mainly concerned with the product X_1 (produced at rate k), which is assumed to be radioactive, decaying in a single step or via a chain of decays to a final stable nuclide. The number of radioactive product atoms N_1 of X_1 present at any time t is the difference between the total number of product atoms formed and the number of radioactive decays that have occurred in time dt. This leads to the differential equation

$$dN_1 = k\,dt - \lambda_1 N_1\,dt \tag{17.6}$$

Integration between the limits of $N_1 = 0$ (this assumes that no radioactive product nuclide X_1 exist at the beginning of the irradiation) and N_1 corresponding to an irradiation time t_{irr}:

$$N_1 = (k/\lambda_1)\left(1 - e^{-\lambda_1 t_{irr}}\right) \tag{17.7}$$

N_1 increases with time towards a maximum value, which is reached when $t_{irr}\lambda_1 \gg 1$ (i.e. $t_{irr} \gg t_{½}$):

$$N_1(\max) = k/\lambda_1 = k\,t_{½}/0.693 \tag{17.8}$$

However, for $t_{irr} \gg t_{½}$, by expanding the exponential into a MacLaurin series, $e^{-\lambda_1 t_{irr}} \approx 1 - \lambda_1\, t_{irr}$, which with (17.7) gives

$$N_1 = k\,t_{irr} \left(\text{for } t_{irr} \ll t_{½}\right) \tag{17.9}$$

If we measure the irradiation time in number of half-lives, $a = t_{irr}/t_{½}$, we obtain

$$N_1 = N_{1,max}\left(1 - 2^{-a}\right) \tag{17.10}$$

This equation shows that irradiation for one half-life ($a = 1$) produces 50% of the maximum amount, two half-lives give 75%, etc.

After the end of the irradiation time, the radioactive product nuclides continue to decay according to their half-lives. We indicate this decay time after the end of irradiation (termed the cooling time) as t_{cool}. The number of product nuclides, which is a function of the irradiation time t_{irr} and cooling time t_{cool}, is given by

$$N_1 = (k/\lambda_1)\left(1 - e^{-0.693 t_{irr}/t_{½}}\right) e^{-0.693 t_{cool}/t_{½}} \tag{17.11}$$

The effect of the cooling time is shown in Figure 17.1 for two sample nuclides.

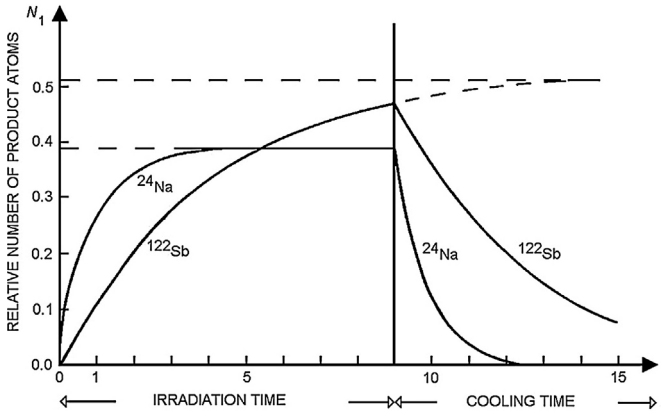


Figure 17.1 Change in number of ^{24}Na ($t_{½}$ 15.0 h) and ^{122}Sb ($t_{½}$ 2.7 d) atoms during and after irradiation.

The production rate k is the number of transformations, N, leading to a number, N_1, of product atoms, divided by the irradiation time, t_{irr}. For a target in a particle flux

$$k = \varphi\, \sigma\, N_t \tag{17.12}$$

where φ is the particle flux (particles of appropriate energy per cm^2 per s), σ the reaction cross section (at the given particle energy, in cm^2) and N_t the number of target atoms with cross section σ; cf. eqn. (11.4). In the case a fission product is produced, $\varphi\, \sigma\, N_t$ is the fission rate, which has to be multiplied by the particular fission yield, y_i, for nuclide i.

If a target is bombarded by a beam of particles the flux is described by a conversion factor and the current I (A) of particles with charge z and the production rate k is

$$k = 6.24 \times 10^{18} I\, \sigma\, N_{\text{v}}\, x\, z^{-1} \tag{17.13}$$

where $N_{\text{v}}\, x$ is the number of target atoms per unit area; cf. eqn. 11.5).

For (17.12) and (17.13) to be valid,

(i) the number of nuclear reactions must remain small compared to the total number of target atoms to justify the assumption of constant N_t or N_{v},

(ii) the projectile energy must be maintained constant in order that the reaction cross-section remains constant, and

(iii) the flux must not be decreased in passing through the target. Depending on the conditions of the irradiation only one or two of these requirements may be valid. Corrections must then be introduced for the non-valid component(s).

Usually the radioactive decay rate A_1 is measured, rather than the number of atoms N_1. Recalling $A = \Delta N/\Delta t = \lambda N$, (17.3) and (17.12) yield

$$A_1 = k\left(1 - \mathrm{e}^{-0.693 t_{\text{irr}}/t_{½}}\right)\mathrm{e}^{-0.693 t_{\text{cool}}/t_{½}} \tag{17.14}$$

This equation represents the basic relationship for the production of radionuclides; in a reactor k is calculated according to (17.12) but for accelerator irradiation, where the current and not the flux is known, (17.13) is normally used.

Let us again take the production of radioactive sodium by reactor irradiation and cyclotron bombardment as examples of the application of these equations. Na_2CO_3 (M = 106) is irradiated in a reactor to produce ^{24}Na, which has a half-life of 14.66 h and emits β^- and γ-rays. The reaction cross-section of ^{23}Na (100% in natural sodium) is 0.53 b for thermal neutrons. For a 5 g sample, 60 h irradiation time, and a thermal flux of 10^{12} n cm^{-2} s^{-1}, the activity at the end of the irradiation time will be:

$$A = 10^{12} \times 0.53 \times 10^{-24} \times 2 \times 5 \times 6.02 \times 10^{23}\left(1 - e^{-0.693 \times 60/15}\right)/106$$
$$= 2.8 \times 10^{10}\,\text{Bq} \tag{17.15}$$

^{24}Na may be produced through accelerator irradiation of magnesium by the reaction ^{26}Mg (d,α) ^{24}Na. With 22 MeV D^+ ions the reaction cross-section is about 25 mb. For a 0.1 mm thick magnesium foil (M = 24.3, $\rho = 1.74$ g cm^{-3}) with an area larger than the projectile beam, and a 2 h irradiation at 100 μA, the activity of ^{24}Na produced (natural magnesium contains 11.0% ^{26}Mg) at end of irradiation is:

$$A = 6.24 \times 10^{18} \times 100 \times 10^{-6} \times 25 \times 10^{-27}\left(0.11 \times (1.74/24.3)\right.$$
$$\left. \times\, 6.02 \times 10^{23} \times 0.01\right) \times \left(1 - e^{-0.693 \times 2/15}\right) = 6.53 \times 10^{7}\text{Bq} \tag{17.16}$$

17.3. SECOND-ORDER REACTIONS

For large cross-sections, short half-lives, and long bombardments, second-order capture products may be formed. If the first-order product is radioactive, then its concentration at any time is dependent on (i) the decay constant, (ii) the cross-section for production of the second-order product, as well as (iii) the cross-section for its own production. These possibilities lead to a scheme such as Figure 17.2. For simplicity, we may assume that the induced nuclear transformations only involve single-neutron capture, and that only β^- decay occurs. However, it should be obvious that the scheme in Figure 17.2, with suitable additions, is applicable to all kinds of nuclear formation and decay reactions.

For the (n,γ) β^- case the upper horizontal row of Figure 17.2 represents the successive formation of higher isotopes of the target element (the constant Z-chain) and the vertical rows the isobaric decay chains of each of these isotopes (the constant A-chains). The first of these two rows is indicated by heavy arrows. Chains which involve both induced transformations and radioactive decay play a central role in theories about the